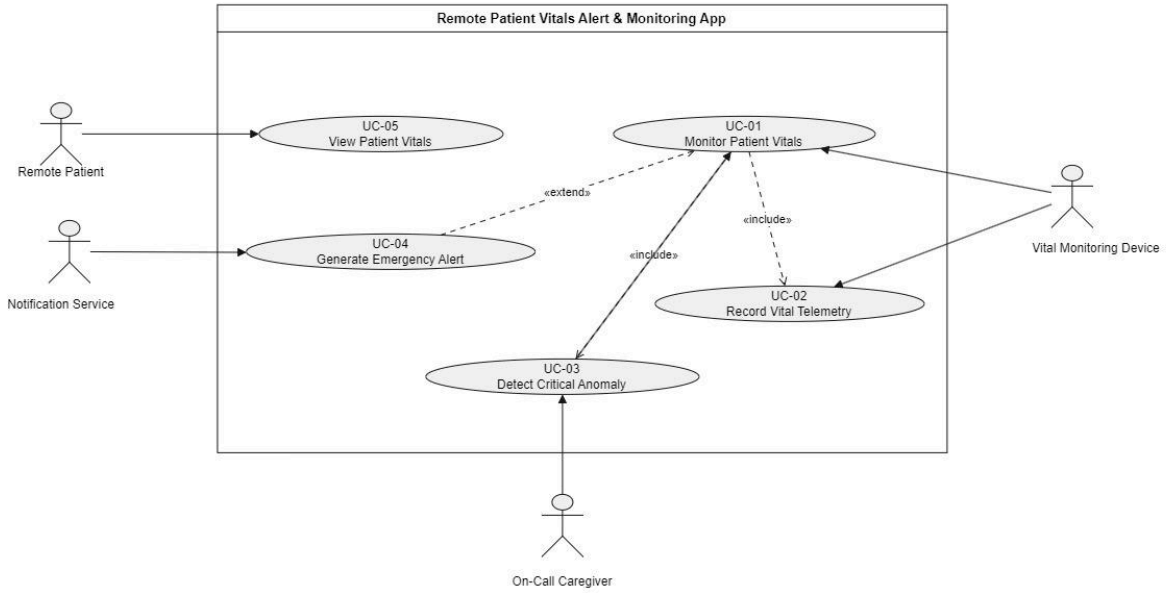
REQUIREMENTS TABLE :-

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: A heart rate above 140 BPM generates a critical anomaly flag within 2 seconds. Fail: The anomaly is ignored or detected after 2 seconds.	Early detection of abnormal vitals is essential for timely medical intervention.
FR-002	Functional	The system shall continuously receive and record the Remote Patient's SpO2, Heart Rate, and BP telemetry data.	High	Pass: All three vital metrics received during a test session are recorded with their timestamp. Fail: Any received metric is lost or recorded without a timestamp.	Continuous telemetry is required for reliable remote patient monitoring.
FR-003	Functional	The system shall generate and send an emergency alert to the appropriate On-Call Caregiver when a critical vital anomaly is detected.	High	Pass: A simulated critical anomaly generates an alert containing the patient identifier, abnormal vital, value, and timestamp. Fail: No alert is generated or required information is missing.	Caregivers must be notified immediately when patient condition becomes critical.
FR-004	Functional	The system shall allow the On-Call Caregiver to view the patient's current and recently recorded vital measurements and alert status.	Medium	Pass: The caregiver can view the latest SpO2, Heart Rate, BP, timestamps, and active alerts. Fail: Any required vital or alert status cannot be displayed.	Provides caregivers with the information needed to assess the patient's condition remotely.
FR-005	Functional	The system shall escalate an unacknowledged critical alert to the next available caregiver in the configured caregiver matrix.	High	Pass: When a critical alert remains unacknowledged for the configured escalation period, the next caregiver receives the alert. Fail: Escalation does not occur after the configured period.	Prevents critical alerts from being missed when the primary caregiver is unavailable.
NFR-001	Nonfunctional - Performance & Availability	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams while maintaining 99.99% availability.	High	Pass: Load testing demonstrates 500 concurrent streams with 99.99% availability and no sustained service failure. Fail: The system cannot maintain the specified load or availability.	Ensures the monitoring system remains reliable as the number of monitored patients increases.
NFR-002	Nonfunctional - Security	The system shall protect patient telemetry and alert data using authenticated access	High	Pass: Unauthenticated users cannot access patient data, and telemetry transmitted between system components	Patient vital information is sensitive and must be protected against unauthorized access.

		and encryption during transmission and storage.		is encrypted. Fail: Patient data can be accessed without authentication or transmitted without encryption.	

Use Case Diagram :-



Use-Case Flow Specification :-

Use Case: UC-04 - Generate Emergency Alert

Main Success Scenario

1. The system continuously receives the patient's SpO2, Heart Rate, and BP telemetry.
2. The system evaluates each received vital measurement against the configured clinical thresholds.
3. The system detects a critical anomaly, such as a heart rate exceeding the configured critical threshold.
4. The system records the abnormal vital value and timestamp.
5. The system generates an emergency alert containing the patient identifier, abnormal vital, measured value, and timestamp.
6. The system sends the emergency alert to the assigned On-Call Caregiver through the notification service.
7. The On-Call Caregiver receives the alert and reviews the patient's current vital information.
8. The caregiver acknowledges the alert.
9. The system records the acknowledgement and marks the alert as acknowledged.
10. The use case ends successfully.

Alternate Flow - Alert Not Acknowledged

Step 8a: The On-Call Caregiver does not acknowledge the emergency alert within the configured escalation period.

Step 8b: The system identifies the alert as unacknowledged.

Step 8c: The system escalates the alert to the next available caregiver in the configured caregiver matrix.

Step 8d: The system records the escalation and sends the emergency notification to the next caregiver.

Step 8e: The use case continues until the alert is acknowledged or the configured escalation procedure is completed.